

(20 Points) Consider a One-Time Pad over $\mathbb{Z}_4 = \{0, 1, \dots, 3\}$ where $C \equiv M + K \pmod{4}$. The key distribution is: $\Pr[K = 0] = 0.1$, $\Pr[K = 1] = 0.3$, and $\Pr[K = 2] = 0.6$ (0 for others). The message distribution is $\Pr[M = 0] = 0.5$, $\Pr[M = 1] = 0.2$, and $\Pr[M = 2] = 0.3$.

- (a) Does this scheme satisfy Perfect Secrecy for the key distribution above?
- (b) Calculate $\Pr[M = 0 \mid C = 1]$.
- (c) Calculate $\Pr[M = 1 \mid C = 3]$.
- (d) Calculate $\Pr[M = 2 \mid C = 2]$.

(20 Points) Consider a One-Time Pad over $\mathbb{Z}_{65536}$ where $C \equiv M + K \pmod{65536}$. The key K is uniform on $\mathbb{Z}_{65536}$. The message distribution is $\Pr[M = 10] = 0.6$, $\Pr[M = 20] = 0.3$, and $\Pr[M = 30] = 0.1$.

- (a) Does this scheme satisfy Perfect Secrecy for the key distribution above?
- (b) Let $a/b := \Pr[C = 30 \mid M = 10]$. Calculate (a, b) .
- (c) Calculate $\Pr[M = 20 \mid C = 20]$.
- (d) Calculate $\Pr[M = 30 \mid C = 10]$.

(10 Points) Consider a symmetric-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ over the message space $\mathcal{M} = \{0, 1\}^n$. The scheme uses a secure PRF $F : \{0, 1\}^n \times \{0, 1\}^4 \rightarrow \{0, 1\}^n$.

The encryption algorithm $\text{Enc}_k(m)$ is defined as:

$$c = (r, F(k, r) \oplus m)$$

where r is sampled uniformly from $\{0, 1\}^4$.

Suppose an adversary $\mathcal{A}$ is allowed to make **4 queries** to the encryption oracle.

- (a) Calculate the probability that at least two of the four queries use the same r .
- (b) Is this encryption scheme CPA-secure?
 - A. Yes.** The proof is by reduction. If there exists a PPT adversary $\mathcal{A}$ that distinguishes c from a truly random string, we can construct a PPT distinguisher $\mathcal{D}$ that breaks the pseudorandomness of F by embedding its own challenge r into the encryption oracle, simulating $c = (r, F(k, r) \oplus m)$ for $\mathcal{A}$.
 - B. Yes.** The scheme is CPA-secure because F is a secure PRF, and the XOR operation ensures that the ciphertext is computationally indistinguishable from random strings for any polynomial number of queries.
 - C. No.** The scheme is insecure because an adversary can use an algebraic attack to break the PRF and recover the secret key k with non-negligible advantage.
 - D. No.** The scheme is insecure because an adversary can compute the XOR sum of two messages when a collision of r occurs.

(10 Points) Consider the encryption scheme $\Sigma^n = (\text{Gen}, \text{Enc}, \text{Dec})$ defined over $\mathbb{Z}_n$:

- $\mathcal{M} = \mathcal{C} = \mathbb{Z}_n \times \mathbb{Z}_n$, $\mathcal{K} = \mathbb{Z}_n^4$.
- Gen : Sample a uniform pair $(k_1, k_2, k_3, k_4) \leftarrow \mathbb{Z}_n^4$.
- Enc : For input $(m_1, m_2) \in \mathbb{Z}_n \times \mathbb{Z}_n$, output $c = (k_3 + m_1 \cdot k_1 \pmod{n}, k_4 + m_2 \cdot k_2 \pmod{n})$.

- (a) Set $n := 8$, and let $a/b := \Pr[M = (6, 7) \mid C = (7, 6)]$. Calculate (a, b) .
- (b) Determine whether the scheme described above is a valid secure encryption scheme.
- A. Yes.** Since the key is chosen uniformly at random from $\mathcal{K}$, the probability $\Pr[C = c \mid M = m]$ is exactly $1/|\mathcal{K}|$, which is independent of the message m .
- B. No.** Due to its correctness does not hold for the entire space.
- C. No.** The encryption function is linear with respect to the message and the key, which allows for algebraic attacks (such as solving systems of linear equations) that can recover the key.
- D. Yes.** The scheme is perfectly secure because the size of the key space $|\mathcal{K}| = n^2$ is strictly larger than the size of the message space $|\mathcal{M}| = n$.

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{3n}$ be a secure PRG with output denoted as $w_1 || w_2 || w_3$, where $w_1, w_2, w_3 \in \{0, 1\}^n$. Bob constructed a PRG $G' : \{0, 1\}^n \rightarrow \{0, 1\}^{3n}$ as follows:

- Obtain $w_1 || w_2 || w_3 := G(s)$.
- Output $(w_1 \oplus w_2) || (w_2 \oplus w_3) || (w_3 \oplus w_1)$.

Is G' a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{A}$ distinguishes G' from random, we can build a distinguisher $\mathcal{B}$ for G that embeds its own challenge z to break G via hybrid argument.
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can break a single instance of G , we can trivially extend this attack by run several G to successfully distinguish the entire output of G' .
- **C. No.** The construction violates the definition of G due to uses it incorrectly with mismatch length.
- **D. No.** The construction allows for a distinguisher with non-negligible advantage in polynomial time.

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a secure PRG. Carol constructed a $G^* : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ defined as follows: on input seed $s \in \{0, 1\}^n$, sample two fresh, independent uniform strings $r_1, r_2 \in \{0, 1\}^{n/2}$, and then output:

$$G^*(s) := G(s \oplus (r_1 || r_2)).$$

Is G^* a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., since $s \oplus (r_1 || r_2)$ is a perfectly uniform seed due to the properties of the one-time pad, any PPT adversary $\mathcal{A}$ that distinguishes $G^*(s)$ from random can be used by a distinguisher $\mathcal{B}$ to break the underlying PRG G by setting the challenge as the input to $\mathcal{A}$.
- **B. Yes.** The proof is by reduction, i.e., assume there exists an attack on the atomic generator G , one can simulate an attack on G^* by fixing $r_1 || r_2 = 0^n$, thereby proving that the security of Carol's construction is strictly lower-bounded by the computational hardness of the original PRG.
- **C. No.** The construction violates the definition. The construction fails to satisfy the definition due to the improper use of randomness.
- **D. No.** The size of the effective seed space allows for a distinguisher with non-negligible advantage in polynomial time. The output $G(s \oplus (r_1 || r_2))$ has a statistical distance from the uniform distribution.

(20 Points) Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a secure PRF. We define a new function $F' : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ as:

$$F'(k, x) = F(k, x) \oplus F(k, x \oplus 1^n \oplus 0^n).$$

The PRF Distinguishing Game:

- **World 0** ($b = 0$): The challenger samples a uniform key $k \leftarrow \{0, 1\}^n$ and computes $y_i = F'(k, x_i)$ for each query ¹.
- **World 1** ($b = 1$): The challenger samples a truly random function $f \leftarrow \text{Funs}[\{0, 1\}^n, \{0, 1\}^n]$ and computes $y_i = f(x_i)$ for each query.

Alice proposes an adversary $\mathcal{A}_{\text{Alice}}$ to interact with the challenger and guess b . For $n = 10$, her strategy is as follows:

- Alice chooses a random $x_0 \in \{0, 1\}^{10}$.
- She constructs 9 additional distinct values $x_1, \dots, x_9$, where $x_i = x_0 \oplus (0^{10-i}1^i)$. (Note: $0^{10-i}1^i$ denotes $10 - i$ zeros followed by i ones. e.g., $x_1 = x_0 \oplus 0000000001$).
- She queries the challenger for the set $\{x_0, x_1, \dots, x_9\}$ and receives $\{y_0, y_1, \dots, y_9\}$.
- If she finds any i, j such that $y_i = y_j$ (for $i \neq j$), she outputs $b' = 0$; otherwise, she outputs $b' = 1$.

Questions:

- Calculate $P_1 = \Pr[b' = 0 \mid b = 1]$, the probability that Alice outputs 0 in World 1.
- Calculate $P_0 = \Pr[b' = 0 \mid b = 0]$, the probability that Alice outputs 0 in World 0.
- Calculate Alice's advantage $Adv = |P_1 - P_0|$.
- Do you think F' is a secure PRF?

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a secure PRG. Alice constructed a $G^{**} : \{0, 1\}^n \rightarrow \{0, 1\}^{16n}$, the output is defined as:

$$G^{**}(x) := G(G(G(G(X)))).$$

Is G^{**} a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{A}$ distinguishes G^{**} from random, we can build a distinguisher $\mathcal{B}$ for G that embeds its own challenge z to break G .
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can break a single instance of G , we can trivially extend this attack by run several G to successfully distinguish the entire output of G^{**} .
- **C. No.** The construction violates the definition of G due to uses it incorrectly with mismatch length.
- **D. No.** The construction allows for a distinguisher with non-negligible advantage in polynomial time.

(5 Points) Let $F : \{0, 1\}^n \times \{0, 1\}^m \rightarrow \{0, 1\}^m$ be a secure PRF. Alice constructs a new function $F' : \{0, 1\}^n \times \{0, 1\}^m \rightarrow \{0, 1\}^m$ defined as:

$$F'(k, x) = F(k, x) \oplus x$$

where $k \in \{0, 1\}^n$ is the secret key and $x \in \{0, 1\}^m$ is the input.

Is F' a secure PRF?

- **A. Yes.** The proof is by reduction. If there exists a PPT adversary $\mathcal{A}$ that distinguishes F' from a random function, we can construct a PPT distinguisher $\mathcal{D}$ that breaks the pseudorandomness of F by simulating the $\oplus x$ operation on F 's oracle outputs.
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ that can break a single instance of F , we can trivially extend this attack by xor it to successfully distinguish the entire output of F' .
- **C. No.** The construction is insecure because x is known to the adversary. By observing $y = F'(k, x)$, an adversary can recover the internal PRF value as $F(k, x) = y \oplus x$, which allows for a successful algebraic attack.
- **D. No.** The construction violates the definition of a PRF because a PRF must be a deterministic function of the key and input.

(5 Points) Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{n+1}$ be a secure PRG. Bob constructed a PRG $G' : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ as follows:

- Sample $s_0 \leftarrow \{0, 1\}^n$.
- For $j = 1, \dots, \ell$: compute $s_j b_j \leftarrow G(s_{j-1})$, where b_j is the last bit.
- Output the string $y = b_1 b_2 \dots b_\ell$.

Is G' a secure PRG?

- **A. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{A}$ distinguishes G' from random, we can build a distinguisher $\mathcal{B}$ for G that embeds its own challenge z to break G via hybrid argument.
- **B. Yes.** The proof is by reduction, i.e., assume there exists a PPT adversary $\mathcal{D}$ can break a single instance of G , we can trivially extend this attack by run several G to successfully distinguish the entire output of G' .
- **C. No.** The construction violates the definition of G due to uses it incorrectly with mismatch length.
- **D. No.** The construction allows for a distinguisher with non-negligible advantage in polynomial time.